

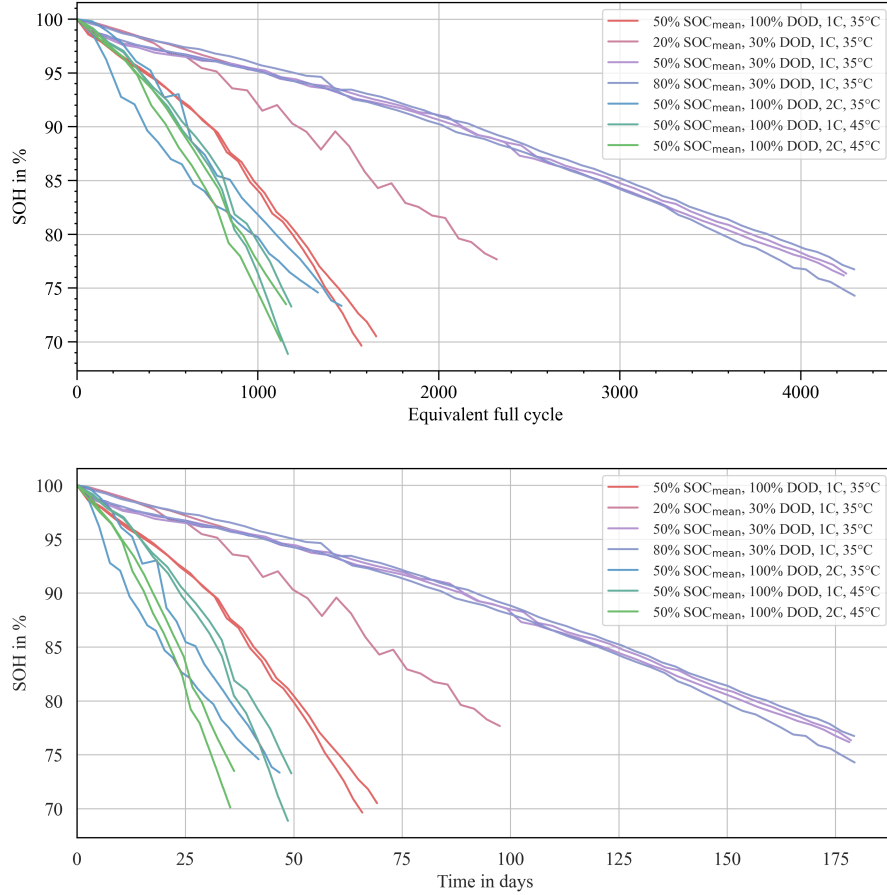


Figure 4.1: SOH progression of the 14 investigated NMC cells: top over equivalent full cycles and bottom over absolute time in days. Colors denote the seven ageing scenarios, with two cells per scenario.

This dataset is used exclusively in Chapter 5. The present section defines the cells, test scenarios, and SOH labels, while the contribution chapter concentrates on the cumulative-current and lag-sequence feature engineering.

4.1.2 LFP Dataset for Robustness Benchmarking and Embedded Deployment

The second dataset is the published LFP DoE Cycle Ageing Dataset recorded at Technische Universität Berlin within the MG-FARM project.[93] It uses commercially available JGCFR18650-1800mAh-3.2V LFP cells with a nominal capacity of 1,8 Ah and a nominal voltage of 3,2V. The ageing campaign follows a three-factor design-of-experiments scheme conducted at a constant ambient temperature of 25 °C. The three varied factors are the charge rate C_{chg} from 0.3 C to 1.0 C, the